

EXPT NO:3	CLEANSE DATA BY HANDLING MISSING VALUES, OUTLIERS, AND INCONSISTENCIES USING TABLEAU AND POWERBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS

1. What are the potential impacts of missing values in a dataset, and what are some common strategies to address them in Tableau and Power BI?

Missing values can reduce the accuracy of data analysis, produce misleading visualizations, and negatively affect predictive models. They may also cause incorrect calculations and incomplete reports. In Tableau and Power BI, missing values can be handled by replacing null values with mean, median, or default values such as "Unknown," removing incomplete records when appropriate, and using calculated fields like **IFNULL()** in Tableau or **IF(ISBLANK())** in Power BI.

2. What is the importance of identifying and addressing outliers in a dataset? How can outliers affect data analysis and model performance?

Outliers are unusually high or low values that differ significantly from the rest of the dataset. They can distort averages, trends, and statistical measures, leading to inaccurate insights and poor machine learning model performance. Identifying outliers helps ensure reliable analysis. They can be managed by filtering, capping extreme values, or flagging them for further investigation.

3. What types of data inconsistencies might you encounter during data cleansing, and how can Tableau and Power BI help in resolving them?

Common inconsistencies include spelling mistakes, duplicate category names, inconsistent capitalization, different date formats, missing labels, and varying currency formats. Tableau provides features such as Groups, Aliases, and Data Interpreter, while Power BI offers Power Query tools like Replace Values, Trim, Clean, Merge, Split, and Change Data Type to standardize and correct inconsistent data.

4. How do you identify patterns in data using visualizations to spot missing values, outliers, and inconsistencies?

Visualizations such as bar charts, histograms, scatter plots, box plots, and tables help identify data quality issues. Missing values appear as blank or null categories, outliers are visible as isolated points in scatter plots or box plots, and inconsistencies become noticeable through unexpected category names or unusual distributions. These visuals make it easier to identify areas requiring data cleaning.

5. What is the role of calculated fields in data cleansing, and how can they be used to handle missing values and outliers in Tableau/Power BI?

Calculated fields automate the cleaning process by replacing missing values, correcting inconsistencies, creating standardized fields, and flagging outliers. Functions such as **IFNULL()**, **ZN()**, and **IF** in Tableau or **IF(ISBLANK())**, **COALESCE()**, and **IF()** in Power BI improve data quality without modifying the original dataset.

IN-LAB EXERCISE:

OBJECTIVE:

To clean a given dataset (e.g., Superstore Sample in Tableau and Financials/Sales Data in Power BI) by identifying and resolving missing values, outliers, and inconsistencies using built-in tools, ensuring the dataset is accurate and ready for analysis.

PROCEDURE:

1. Open the Tool and Connect Dataset

- Open Tableau Desktop , Power BI Desktop.
- Load the built-in dataset (Sample - Superstore in Tableau / Financials in Power BI).
- Preview the dataset to understand its structure.

2. Initial Inspection

- Explore fields such as Sales, Profit, Region, or Product Category.
- Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.

3. Handling Missing Values

- Use Data Interpreter (Tableau) / Power Query (Power BI) to highlight nulls.
- Replace missing numerical values with mean/median, and text with "Unknown".
- Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.

4. Detecting and Managing Outliers

- Create scatter or box plots for Sales and Profit to identify extreme values.
- Choose strategies:
 - Filter them temporarily.
 - Apply caps using calculated columns.
 - Flag them with labels for review.

5. Resolving Inconsistencies

- Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.
- Normalize data formats, such as date fields or currency units.

6. Validation with Visuals

- Build dashboards with histograms, bar charts, and box plots to visually confirm improvements.
- Compare before/after cleaning stages for accuracy.

7. Save the Cleaned Dataset

- Save as .twbx (Tableau) / .pbix (Power BI).
- Optionally export cleaned data to .csv.

8. Summary Dashboard

- Create a visual summary of the cleaning process, highlighting:
 - Fields corrected.
 - Nulls/outliers removed or adjusted.
 - Clean dataset metrics (record count, completeness %).

Experiment Dataset:

1. Tableau: *Sample - Superstore*
2. Power BI: *Financials* or *Sales Data*

How to Access:

Tableau:

- **Navigate to:** *Start Page → Sample Workbooks*
- Or: *Help → Sample Data → Open Sample Data Folder*

Power BI:

- **Navigate to:** *Home → Get Data → Excel → Choose sample datasets (e.g., Financial Sample.xlsx)*
- Microsoft also hosts Power BI sample datasets online.

Fields Used:

1. Sales
2. Profit
3. Region
4. Category
5. Order Date / Transaction Date

Tableau - Book1

File Data Server Window Help

Home Back Forward Refresh Save

Connections Add

- Global_Superstore2
Text file

Files

- Use Data Interpreter
Data Interpreter might be able to clean your Text File workbook.
- Global_Superstore2.csv

New Union
New Table Extension

Global_Superstore2

Connection: Live Extract Filters: 0 Add

Global_Superstore2.csv 24 fields 51290 rows 100 rows

Name: Global_Superstore2.csv Description: No description available.

Type	Field Name	Physical Table	Rem...
#	Row ID	Global_Superstore2...	Row ID
Abs	Order ID	Global_Superstore2...	Order ...
Cal	Order Date	Global_Superstore2...	Order ...
Cal	Ship Date	Global_Superstore2...	Ship D...
Abs	Ship Mode	Global_Superstore2...	Ship ...
Abs	Customer ID	Global_Superstore2...	Custo...
Abs	Customer Name	Global_Superstore2...	Custo...
Cal	Segment	Global_Superstore2...	Segm...
Cal	City	Global_Superstore2...	City
Cal	State	Global_Superstore2...	State
Cal	Country	Global_Superstore2...	Country
Cal	Postal Code	Global_Superstore2...	Posta...

#	Abs	Cal	Cal	Abs	Abs
Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID
38540	CA-2014-118892	18-08-2014	23-08-2014	Second Class	TP-21415
18241	ES-2013-1434123	29-01-2013	05-02-2013	Standard Class	JB-36000
23013	ID-2012-78207	18-09-2012	21-09-2012	Second Class	AM-10705
45616	MO-2013-8630	23-11-2013	23-11-2013	Same Day	AB-255
25026	IN-2014-64774	13-06-2012	13-06-2012	Same Day	RP-19270
27962	IN-2013-48765	28-10-2013	31-10-2013	First Class	BK-11260
26669	IN-2013-69730	12-12-2013	14-12-2013	First Class	JM-15250
29600	IN-2012-48240	25-05-2012	28-05-2012	Second Class	GP-14740
38411	US-2013-102239	06-05-2013	07-05-2013	First Class	LW-16990
19195	ES-2013-3467296	02-12-2013	02-12-2013	Same Day	NZ-18565
46366	CA-2014-6550	23-12-2014	25-12-2014	Second Class	SV-10815
26731	ID-2013-25742	28-01-2013	30-01-2013	First Class	SC-20695
31278	IN-2012-86698	07-05-2012	07-05-2012	Same Day	NC-18625
12035	ES-2013-2903666	27-11-2013	29-11-2013	First Class	NF-18385
29149	IN-2012-30446	28-11-2012	02-12-2012	Second Class	NZ-18565
15896	IT-2012-1779015	29-09-2012	29-09-2012	Same Day	PM-18940
18307	ES-2013-3903130	03-09-2013	05-09-2013	Second Class	MZ-17335

Go to Worksheet

Data Source Sheet 1

#	Abc_Global_Superstore2.csv	Abc_Global_Superstore2.csv	Abc_Global_Superstore2.csv	Abc_Global_Superstore2.csv	Abc_Global_Superstore2.csv	
Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	
38540	CA-2014-118892	18-08-2014	23-08-2014	Second Class	TP-21415	T
18241	ES-2013-1434123	29-01-2013	05-02-2013	Standard Class	JB-16000	J
23013	ID-2012-78207	18-09-2012	21-09-2012	Second Class	AM-10705	A
45616	MO-2013-8630	23-11-2013	23-11-2013	Same Day	AB-255	A
25026	IN-2012-64774	13-06-2012	13-06-2012	Same Day	RP-19270	R
27962	IN-2013-48765	28-10-2013	31-10-2013	First Class	BK-11260	B
26669	IN-2013-69730	12-12-2013	14-12-2013	First Class	JM-15250	J
29600	IN-2012-48240	25-05-2012	28-05-2012	Second Class	GP-14740	G
38411	US-2013-102239	06-05-2013	07-05-2013	First Class	LW-16990	L
19195	ES-2013-3467296	02-12-2013	02-12-2013	Same Day	NZ-18565	N
46366	CA-2014-6550	23-12-2014	25-12-2014	Second Class	SV-10815	S
26731	ID-2013-25742	28-01-2013	30-01-2013	First Class	SC-20695	S
31278	IN-2012-86698	07-05-2012	07-05-2012	Same Day	NC-18625	N
12035	ES-2013-2903666	27-11-2013	29-11-2013	First Class	NF-18385	N
29149	IN-2012-30446	28-11-2012	02-12-2012	Second Class	NZ-18565	N
15896	IT-2012-1779015	29-09-2012	29-09-2012	Same Day	PM-18940	P
18307	ES-2013-3903130	03-09-2013	05-09-2013	Second Class	MZ-17335	M

Tableau - Book1

File Edit Server Window Help

Global_Superstore2

Connections Add

Global_Superstore2
Text file

Global_Superstore2.csv 24 fields 51290 rows

100 rows

Files

- Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.
- Global_Superstore2.csv
- New Union
- New Table Extension

Name: Global_Superstore2.csv

Description: No description available.

Fields

Type	Field Name	Physical Table	Rem...
#	Row ID	Global_Superstore2...	Row ID
ABC	Order ID	Global_Superstore2...	Order ...
📅	Order Date	Global_Superstore2...	Order ...
📅	Ship Date	Global_Superstore2...	Ship D...
ABC	Ship Mode	Global_Superstore2...	Ship ...
ABC	Customer ID	Global_Superstore2...	Custo...
ABC	Customer Name	Global_Superstore2...	Custo...
ABC	Segment	Global_Superstore2...	Segm...
🌐	City	Global_Superstore2...	City
🌐	State	Global_Superstore2...	State
🌐	Country	Global_Superstore2...	Country
📮	Postal Code	Global_Superstore2...	Postal...

#	ABC	📅	📅	ABC	ABC
Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID
32298	CA-2012-124891	31-07-2012	31-07-2012	Same Day	RH-19495
26341	IN-2013-77878	05-02-2013	07-02-2013	Second Class	JR-16210
25330	IN-2013-71249	17-10-2013	18-10-2013	First Class	CR-12730
13524	ES-2013-1579342	28-01-2013	30-01-2013	First Class	KM-16375
47221	SG-2013-4320	05-11-2013	06-11-2013	Same Day	RH-9495
22732	IN-2013-42360	28-06-2013	01-07-2013	Second Class	JM-15655
30570	IN-2011-81826	07-11-2011	09-11-2011	First Class	TS-21340
31192	IN-2012-86369	14-04-2012	18-04-2012	Standard Class	MB-18085
40155	CA-2014-135909	14-10-2014	21-10-2014	Standard Class	JW-15220
40936	CA-2012-116638	28-01-2012	31-01-2012	Second Class	JH-15985
34577	CA-2011-102988	05-04-2011	09-04-2011	Second Class	GM-14695
28879	ID-2012-28402	19-04-2012	22-04-2012	First Class	AJ-10780
45794	SA-2011-1830	27-12-2011	29-12-2011	Second Class	MM-7260
4132	MX-2012-130015	13-11-2012	13-11-2012	Same Day	VF-21715
27704	IN-2013-73951	06-06-2013	08-06-2013	Second Class	PF-19120
13779	ES-2014-5099955	31-07-2014	03-08-2014	Second Class	BP-11185
36178	CA-2014-143567	03-11-2014	06-11-2014	Second Class	TB-21175

Go to Worksheet x

#	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	
32298	CA-2012-124891	31-07-2012	31-07-2012	Same Day	RH-19495	
26341	IN-2013-77878	05-02-2013	07-02-2013	Second Class	JR-16210	
25330	IN-2013-71249	17-10-2013	18-10-2013	First Class	CR-12730	
13524	ES-2013-1579342	28-01-2013	30-01-2013	First Class	KM-16375	
47221	SG-2013-4320	05-11-2013	06-11-2013	Same Day	RH-9495	
22732	IN-2013-42360	28-06-2013	01-07-2013	Second Class	JM-15655	
30570	IN-2011-81826	07-11-2011	09-11-2011	First Class	TS-21340	
31192	IN-2012-86369	14-04-2012	18-04-2012	Standard Class	MB-18085	
40155	CA-2014-135909	14-10-2014	21-10-2014	Standard Class	JW-15220	
40936	CA-2012-116638	28-01-2012	31-01-2012	Second Class	JH-15985	
34577	CA-2011-102988	05-04-2011	09-04-2011	Second Class	GM-14695	
28879	ID-2012-28402	19-04-2012	22-04-2012	First Class	AJ-10780	
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4132	MX-2012-130015	13-11-2012	13-11-2012	Same Day	VF-21715	
27704	IN-2013-73951	06-06-2013	08-06-2013	Second Class	PF-19120	
13779	ES-2014-5099955	31-07-2014	03-08-2014	Second Class	BP-11185	
36178	CA-2014-143567	03-11-2014	06-11-2014	Second Class	TR-21175	

Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Home Left Arrow Right Arrow Undo Redo Refresh Sort Filter Fit Width Show Me

Data Analytics Pages Columns Rows

Global_Superstore2 Postal Code Clean

Search

Tables

- Category
- City
- Country
- Customer ID
- Customer Name
- Market
- Order Date
- Order ID
- Order Priority
- Postal Code
- Postal Code Clean**
- Product ID
- Product Name
- Region
- Row ID
- Segment
- Ship Date
- Ship Mode
- State
- Sub-Category
- Measure Names
- Discount
- Profit
- Quantity
- Sales
- Shipping Cost
- Global_Superstore2.csv...

Filters

Marks

Automatic

Color Size Text Detail Tooltip

Sheet 1

Postal C...	
0	Abc
1040	Abc
1453	Abc
1752	Abc
1810	Abc
1841	Abc
1852	Abc
1915	Abc
2038	Abc
2138	Abc
2148	Abc
2149	Abc
2151	Abc
2169	Abc
2740	Abc
2886	Abc
2895	Abc
2908	Abc
2920	Abc
3060	Abc
3301	Abc
3820	Abc
4240	Abc
4401	Abc
5408	Abc
6010	Abc
6040	Abc
6360	Abc
6450	Abc
6457	Abc
6460	Abc
6484	Abc
6708	Abc
6810	Abc

Data Source Sheet 1 632 marks, 632 rows by 1 column

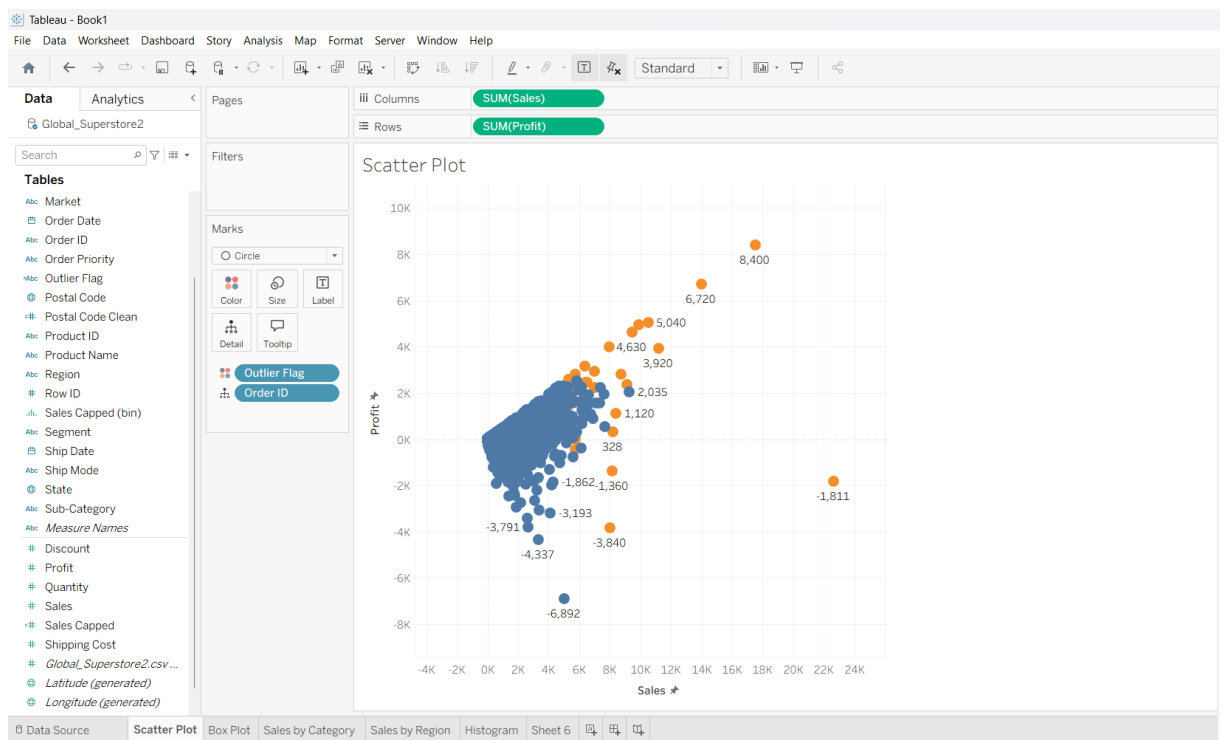
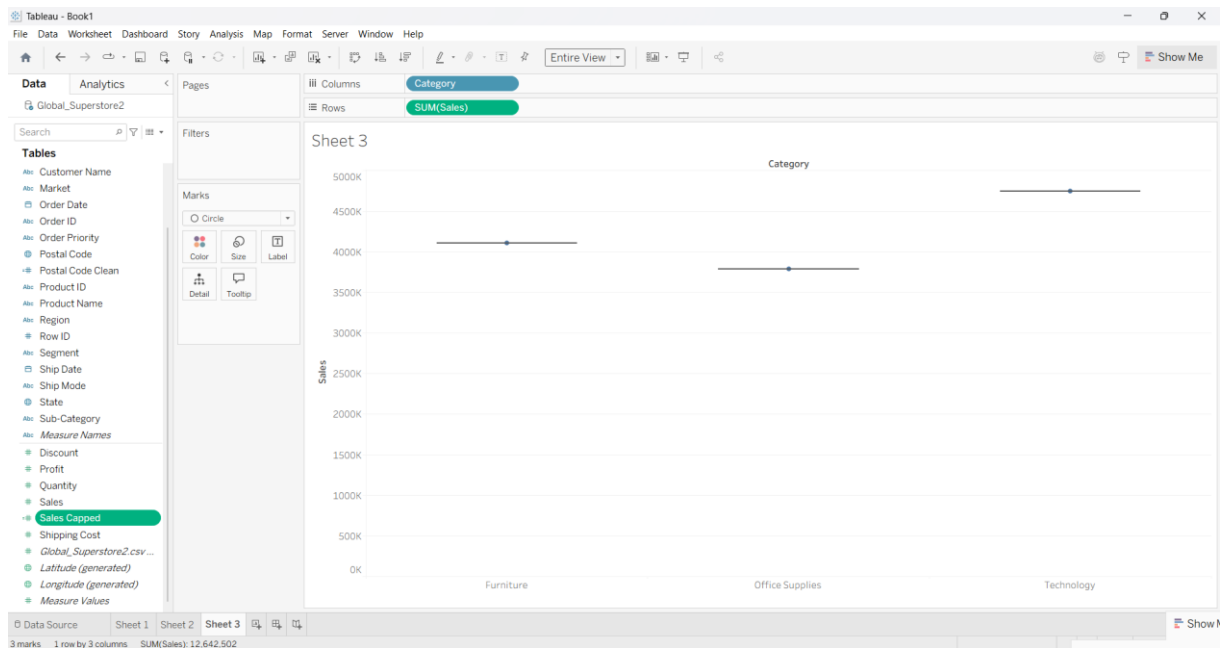
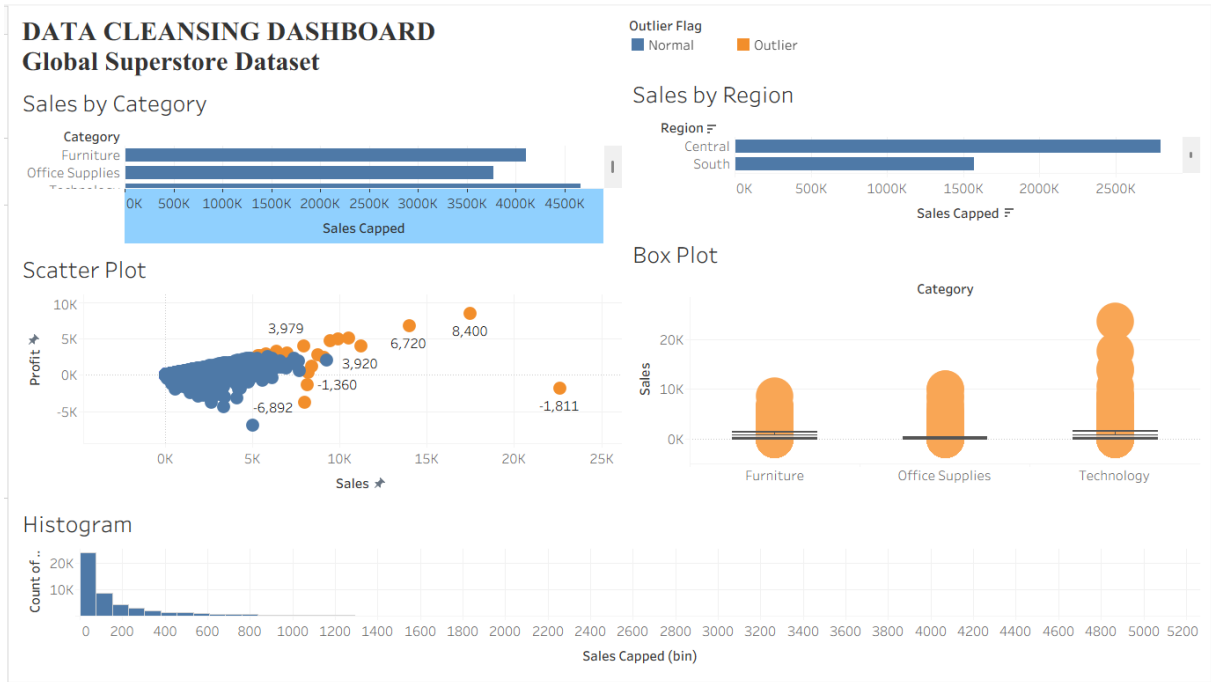


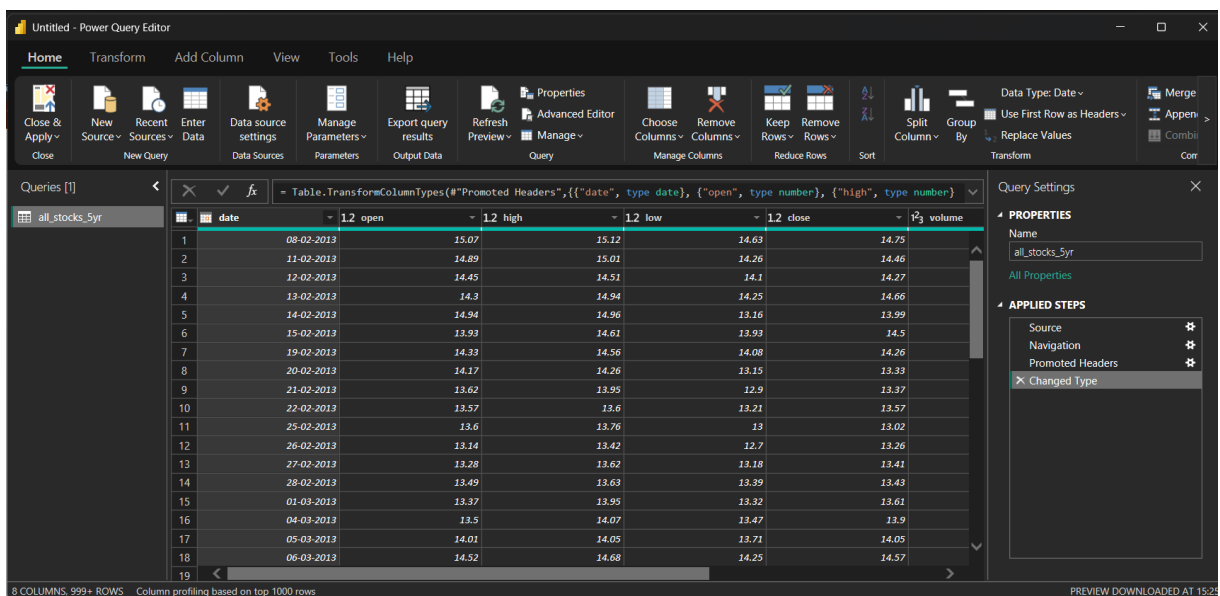
TABLEAU DASHBOARD :



Observation

- The Global Superstore dataset was successfully imported into Tableau.
- Missing values were found in the **Postal Code** field.
- Missing values were handled using the IFNULL() calculated field.
- Outliers were identified using **Scatter Plot** and **Box-and-Whisker Plot**.
- Sales values were capped using a calculated field for analysis.
- Categorical fields were checked for inconsistencies and standardized where necessary.
- The cleaned dataset was validated using visualizations and summarized in a dashboard.

POWER BI Cleaning process :



Untitled - Power Query Editor

Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows Table

Transpose Reverse Rows

Data Type: Decimal Number Detect Data Type

Replace Values Fill Move Pivot Column Convert to List Any Column

Unpivot Columns Split Column Format Extract Parse Text Column

Merge Columns Statistics Standard Scientific Information

Trigonometry Date Time Duration Date & Time Column

Queries [1]

all_stocks_Syr

Table.ReplaceValue(#"Filtered Rows",null,0,Replacer.ReplaceValue,{"high"})

1 08-02-2013 15.07 15.12 14.63 14.75 8407500

2 11-02-2013 14.89 15.01 14.26 14.46 8882000

3 12-02-2013 14.45 14.51 14.1 14.27 8126000

4 13-02-2013 14.3 14.94 14.25 14.66 10259500

5 14-02-2013 14.94 14.96 13.16 13.99 31879900

6 15-02-2013 13.93 14.61 13.93 14.5 15628000

7 19-02-2013 14.33 14.56 14.08 14.26 11354400

8 20-02-2013 14.17 14.26 13.15 13.33 14725200

9 21-02-2013 13.62 13.95 12.9 13.37 11922100

10 22-02-2013 13.57 13.6 13.21 13.57 6071400

11 25-02-2013 13.6 13.76 13 13.02 7186400

12 26-02-2013 13.14 13.42 12.7 13.26 9419000

13 27-02-2013 13.28 13.62 13.18 13.41 7390500

14 28-02-2013 13.49 13.63 13.39 13.43 6143600

15 01-03-2013 13.37 13.95 13.32 13.61 7376800

16 04-03-2013 13.5 14.07 13.47 13.9 8174800

17 05-03-2013 14.01 14.05 13.71 14.05 7676100

18 06-03-2013 14.52 14.68 14.25 14.57 13243200

19

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Table.Distinct(#"Replaced Value3")

1 date 08-02-2013 15.07 15.12 14.63 14.75 8407500 AAL

2 11-02-2013 14.89 15.01 14.26 14.46 8882000 AAL

3 12-02-2013 14.45 14.51 14.1 14.27 8126000 AAL

4 13-02-2013 14.3 14.94 14.25 14.66 10259500 AAL

5 14-02-2013 14.94 14.96 13.16 13.99 31879900 AAL

6 15-02-2013 13.93 14.61 13.93 14.5 15628000 AAL

7 19-02-2013 14.33 14.56 14.08 14.26 11354400 AAL

8 20-02-2013 14.17 14.26 13.15 13.33 14725200 AAL

9 21-02-2013 13.62 13.95 12.9 13.37 11922100 AAL

10 22-02-2013 13.57 13.6 13.21 13.57 6071400 AAL

11 25-02-2013 13.6 13.76 13 13.02 7186400 AAL

12 26-02-2013 13.14 13.42 12.7 13.26 9419000 AAL

13 27-02-2013 13.28 13.62 13.18 13.41 7390500 AAL

14 28-02-2013 13.49 13.63 13.39 13.43 6143600 AAL

15 01-03-2013 13.37 13.95 13.32 13.61 7376800 AAL

16 04-03-2013 13.5 14.07 13.47 13.9 8174800 AAL

17 05-03-2013 14.01 14.05 13.71 14.05 7676100 AAL

18 06-03-2013 14.52 14.68 14.25 14.57 13243200 AAL

19 07-03-2013 14.7 14.93 14.5 14.82 9125300 AAL

20 08-03-2013 14.99 15.2 14.84 14.92 10593700 AAL

21 11-03-2013 14.85 15.15 14.71 15.13 6961800 AAL

22 12-03-2013 15.14 15.6 14.95 15.5 8999100 AAL

23 13-03-2013 15.54 16.2 15.48 15.91 11380000 AAL

24 14-03-2013 15.98 16.36 15.93 16.25 8383300 AAL

25 15-03-2013 16.45 16.54 15.88 15.98 17667700 AAL

26 16-03-2013 16.9 16.77 16.71 16.76 6514700 AAL

PREVIEW DOWNLOADED AT 15:25

Query Settings

PROPERTIES

Name

all_stocks_Syr

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Filtered Rows

Replaced Value

Query Settings

PROPERTIES

Name

all_stocks_Syr

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Filtered Rows

Replaced Value

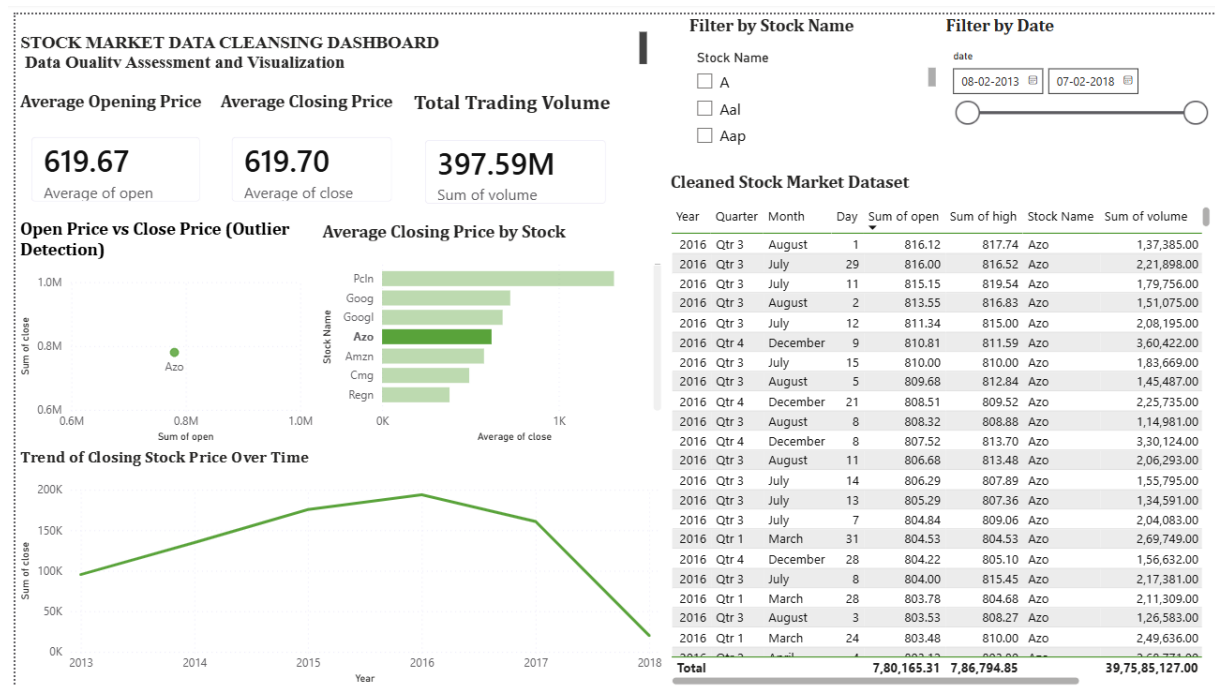
Replaced Value1

Replaced Value2

Replaced Value3

Removed Duplicates

POWER BI DASHBOARD :



POWER BI DASHBOARD LINK :

https://kumaragurudtsteam-my.sharepoint.com/:u:/r/personal/roshniangel_24ad_kct_ac_in/Documents/English/EX%203%20POWER%20BI.pbix?csf=1&web=1&e=wW54lc

Data Cleaning Summary :

- Dataset imported successfully
- Missing values identified and handled
- Data types verified
- Duplicate records removed
- Text values standardized
- Outliers identified using Scatter Plot
- Dataset validated successfully

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What method did you use to replace missing values in numeric fields?

Missing values in numeric fields were replaced using calculated fields and data transformation tools. In Tableau, the IFNULL() function was used to replace null values with **0**. In Power BI, **Power Query** and the **Replace Values** option were used to replace null values with **0**, ensuring that the dataset was complete and ready for analysis.

2. How were outliers handled in your dataset and what visual confirmed their correction? Outliers were identified using **Scatter Plot** and **Box-and-Whisker Plot** in Tableau, and the **Scatter Chart** in Power BI. The outliers were managed by creating a calculated field to cap extreme values and by flagging them for analysis. The visualizations confirmed the presence and handling of outliers, making the data more suitable for analysis.

3. What inconsistencies were detected in categorical values and how were they standardized?

Categorical fields such as **Region**, **Category**, and **Stock Name** were checked for inconsistencies like spelling differences, extra spaces, and formatting issues. In Tableau, **Aliases** and **Groups** were used where required, while in Power BI, **Trim**, **Clean**, and **Replace Values** were applied to standardize the text values. No major inconsistencies were found in the given datasets after verification.

4. Describe how calculated fields helped in resolving data issues.

Calculated fields helped replace missing values, cap extreme values, and flag outliers for analysis. In Tableau, functions such as IFNULL() and conditional IF statements were used, while in Power BI, calculated columns and DAX expressions were used to improve data quality. These calculations ensured consistent and reliable data for visualization.

5. How did your dashboard demonstrate the effectiveness of the data cleaning process?

The dashboard demonstrated the effectiveness of the data cleaning process by presenting clean and accurate visualizations, including bar charts, line charts, scatter plots, box plots, histograms, and summary cards. It highlighted the handling of missing values, identification of outliers, standardized data, and key metrics, confirming that the dataset was successfully cleaned and ready for further analysis.

RESULT:

The dataset was successfully cleaned and prepared for accurate analysis using Tableau and Power BI. Issues such as nulls, inconsistencies, and outliers were identified and resolved, improving data quality significantly.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		